

Year 11

Mathematics

Assessment Task 3

2018

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- In Questions 11 - 14, show relevant mathematical reasoning and/or calculations
- A reference sheet has been provided

Total marks – 70

Section I

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section.

Section II

60 marks

- Attempt Questions 11-14
- Start each question in a new writing booklet
- Write your name on the front cover of each writing booklet
- If you do not attempt a question, submit a blank booklet marked with your name and "N/A" on the front cover
- Allow about 1 hour 45 minutes for this section

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Section I**10 marks****Attempt Questions 1-10**

Use the multiple choice answer sheet for Questions 1-10.

1 Which of the following is the expansion of $(2a - b)^2$?

(A) $4a^2 - b^2$

(B) $4a^2 - 4ab + b^2$

(C) $4a^2 + 4ab + b^2$

(D) $2a^2 - 2ab + b^2$

2 Calculate $\sqrt{9.96 \times 10^{-4}}$ correct to 2 significant figures.

(A) 31.5×10^{-3}

(B) 3.2×10^{-2}

(C) 31.6×10^{-3}

(D) 0.03

3 Simplify the expression $\sqrt{12} - \sqrt{50} + \sqrt{3} - \sqrt{8}$.

(A) $3\sqrt{3} - 3\sqrt{2}$

(B) $3\sqrt{3} - 7\sqrt{2}$

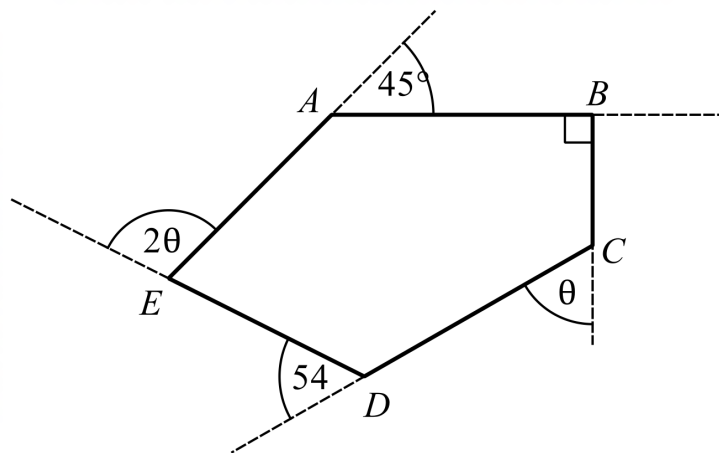
(C) $\sqrt{43}$

(D) $-5\sqrt{6}$

4 What are the solutions to the inequality $|2x - 1| > 5$?

- (A) $-2 < x < 3$
- (B) $-3 < x < 3$
- (C) $x < -3$ or $x > 3$
- (D) $x < -2$ or $x > 3$

5 The diagram shows an irregular pentagon. Some of the interior and exterior angle sizes have been marked on the diagram.

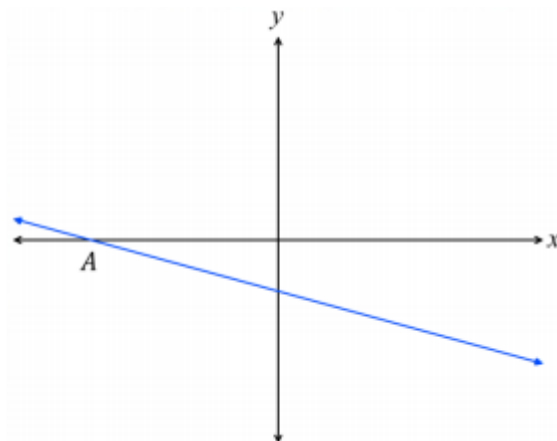


NOT TO SCALE

Find the value of θ .

- (A) 171°
- (B) 87°
- (C) 56°
- (D) 57°

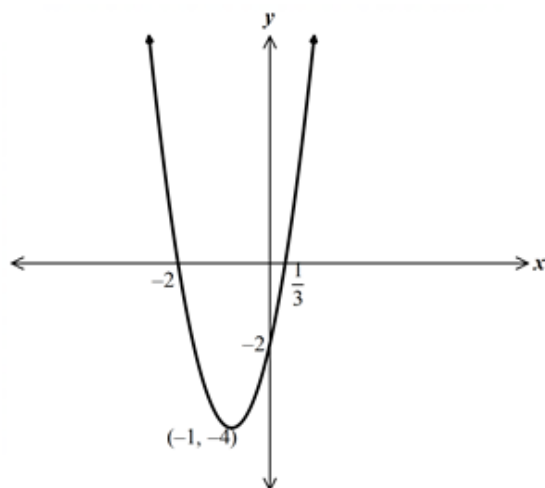
- 6 The line $2x + 4y + 8 = 0$ cuts the x-axis at A .



The coordinates of A are:

- (A) $(-4, 0)$
 - (B) $(0, -4)$
 - (C) $(-2, 0)$
 - (D) $(0, -2)$
- 7 The domain of the function $f(x) = \sqrt{9 - x^2}$ is:
- (A) $-9 \leq x \leq 9$
 - (B) $-3 \leq x \leq 3$
 - (C) $0 \leq y \leq 3$
 - (D) $x \geq 0$
- 8 Which of the following is an expression for $\frac{1 - \sin^2 x}{\cot x}$?
- (A) $\cos^2 x$
 - (B) $\tan x$
 - (C) $\sin x \cos x$
 - (D) $\tan x - \sin x \cos x$

- 9 Which of the equations below describes the parabola in the diagram?



- (A) $y = (3x - 1)(x + 2)$
(B) $y = (3x + 1)(x - 2)$
(C) $y = (x - 3)(x + 2)$
(D) $y = (x - 1)(x - 4)$

- 10 Find $\frac{d}{dx}(3x^4 - 7x^{-3})$

- (A) $3x^3 - 7x^{-2}$
(B) $12x^3 - \frac{21}{x^2}$
(C) $12x^3 + \frac{21}{x^4}$
(D) $12x^3 - \frac{21}{x^4}$

End of Section I

Section II
60 marks
Attempt Questions 11-14

Answer each question in separate writing booklets. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

Marks

(a) Solve each of the following equations.

(i) $2^{x-1} = 4$ **1**

(ii) $2x^2 + 7x + 3 = 0$ **2**

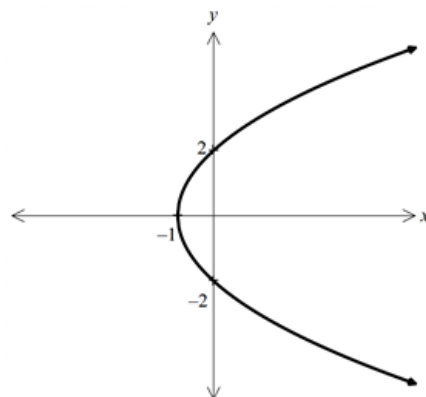
(iii) $|4x + 3| = 5x$ **3**

(b) Express with a rational denominator $\frac{\sqrt{2}}{2 + \sqrt{3}}$ **2**

(c) Solve $2 \sin \theta = -1$ in the domain $0^\circ \leq \theta \leq 360^\circ$ **2**

(d) Prove that $\tan \theta \operatorname{cosec} \theta = \sec \theta$ **2**

(e) For the graph shown below:



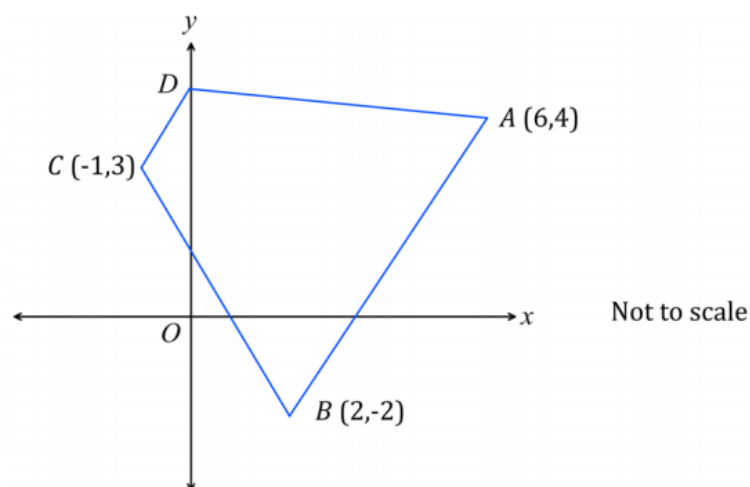
(i) Is this a graph of a function? Justify your answer. **1**

(ii) State the domain and range. **2**

END OF QUESTION 11

Question 12 (15 marks)**Marks**

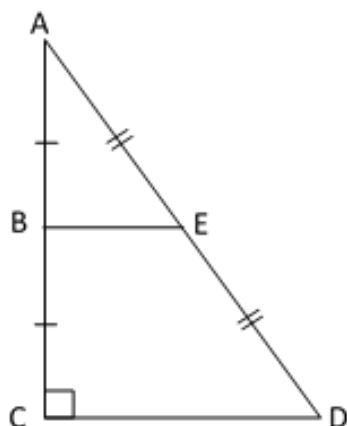
- (a) The points $A(6, 4)$, $B(2, -2)$ and $C(-1, 3)$ are plotted on a number plane. The point D lies on the y -axis such that AB is parallel to CD .



- | | | |
|-------|---|----------|
| (i) | Find the exact length of AB . | 1 |
| (ii) | Find the gradient of AB . | 1 |
| (iii) | Show that the equation of the line AB is $3x - 2y - 10 = 0$ | 2 |
| (vi) | Find the perpendicular distance from point C to AB . | 2 |
- (b) State the equation of a circle with centre $(0, 2)$ and radius of 3 units. **2**
- (c) For the function $f(x) = \begin{cases} -2 & \text{for } x < 0 \\ x^3 & \text{for } x \geq 0 \end{cases}$, find $f(-1) - f(2)$. **2**

QUESTION 12 CONTINUES ON PAGE 10

- (d) The diagram shows $\triangle ACD$, right angled at C . B and E are the midpoints of AC and AD , respectively.



- | | | |
|-------|---|----------|
| (i) | Prove that $\triangle ACD$ and $\triangle ABE$ are similar. | 2 |
| (ii) | Hence, or otherwise, prove that BE is parallel to CD | 1 |
| (iii) | If $AB = 4$ cm, $BE = 3$ cm, and $AE = 5$ cm, find the area of the trapezium $BCDE$. | 2 |

END OF QUESTION 12

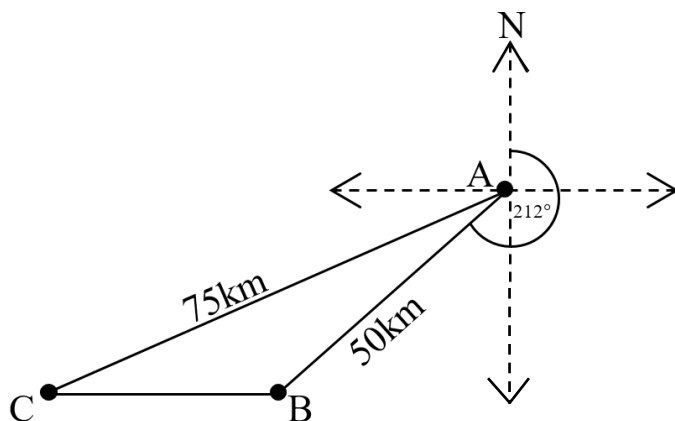
Question 13 (15 marks)**Marks**

- (a) A, B, C are 3 co-linear points of which, A is the point $(-3, 2)$ and $B (2, 4)$ is the midpoint of AC . Find the coordinates of C . **2**
- (b) Given the function $f(x) = x^2 - 7x + 6$
- (i) Find $f(x+h)$ **1**
- (ii) Find $f(x+h) - f(x)$ **1**
- (iii) Hence show that $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = 2x - 7$ **1**
- (c) Evaluate $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$ **2**
- (d) Differentiate the following.
- (i) $6x^2 - x + 2$ **1**
- (ii) $3\sqrt{x}$ **1**
- (iii) $(x^4 - 2x)^3$ **2**
- (iv) $x^2(x-4)^5$ **2**
- (v) $\frac{2x}{7-3x}$ **2**

END OF QUESTION 13

Question 14 (15 marks)**Marks**

- (a) Find the equation of the normal to the curve $y = 3x^2 - 2$ at the point on the curve where $x = 1$. **3**
- (b) A ship sailed 50 kilometres from point A to point B on a bearing of 212° . From point B , it sailed due west to reach point C . The distance AC is 75 kilometres.



- (i) Show that $\sin C = 50 \times \frac{\sin 122^\circ}{75}$. **3**
- (ii) Hence, find the bearing of point C from point A . **2**
(Give your answer correct to the nearest degree.)
- (c) For the equations $y = |x + 1|$ and $y = -x^2 + x + 2$,
- (i) Sketch the graph of $y = |x + 1|$ in your writing booklet. Show all important features. **2**
- (ii) On the same axes as (i), sketch the graph of $y = -x^2 + x + 2$. Show where the curve cuts the x and y axes and the coordinates of its vertex. **3**
- (iii) Shade the region bound by $y < |x + 1|$ and $y < -x^2 + x + 2$. **2**

END OF EXAMINATION

Student Name:

2018 Year 11 Assessment Task 3

Mathematics

Section I Multiple-Choice Answer Sheet

- | | | | | |
|-----------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☐ | D ☐ |

Question 11

(a). (i) $2^{x-1} = 4$
 $2^{x-1} = 2^2$
 $x-1 = 2$
 $x = 3$

①

(ii) $2x^2 + 7x + 3 = 0$

$p: 6 \quad s: 7 \quad f: (6, 1)$

$$2x^2 + 6x + x + 3 = 0$$

$$2x(x+3) + (x+3) = 0$$

$$(x+3)(2x+1) = 0$$

$$x+3 = 0$$

$$x = -3$$

①

$$2x+1 = 0$$

$$x = -\frac{1}{2}$$

①

(iii) $|4x+3| = 5x$

Case 1: $4x+3 > 0$

$$4x+3 = 5x$$

$$x = 3$$

①

Test

$$|4 \times 3 + 3| = 5 \times 3$$

$$15 = 15$$

$$LHS = RHS$$

Case 2: $4x+3 < 0$

$$-4x-3 = 5x$$

$$-3 = 9x$$

$$x = -\frac{1}{3}$$

①

Test

$$|4x - \frac{1}{3} + 3| = 5x - \frac{1}{3}$$

$$\frac{5}{3} \neq -\frac{5}{3}$$

$$LHS \neq RHS$$

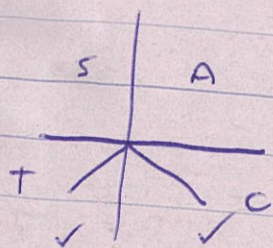
$$x = 3 \quad \text{①}$$

$$(b) \frac{\sqrt{2}}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} \quad (i) = \frac{\sqrt{2}(2-\sqrt{3})}{4-3}$$

$$= 2\sqrt{2} - \sqrt{6} \quad (i)$$

$$(c) 2 \sin \theta = -1$$

$$\sin \theta = \frac{-1}{2}$$



Basic angle = 30°

In Q_3

$$\theta = 180 + 30$$

$$\theta = 210^\circ \quad (i)$$

In Q_4

$$\theta = 360 - 30$$

$$\theta = 330^\circ \quad (i)$$

$$\therefore \theta = 210^\circ, 330^\circ$$

$$(d) \tan \theta \operatorname{cosec} \theta = \sec \theta$$

LHS:

$$\frac{\sin \theta}{\cos \theta} \times \frac{1}{\sin \theta} \quad (i)$$

$$\frac{1}{\cos \theta}$$

$$\sec \theta \quad (i)$$

$$\therefore \text{LHS} = \text{RHS}$$

(e)(i) No, does not pass straight line test.
More than one y -value for each x -value. (1)

(ii) Domain $\{x \geq -1, \text{ all real } x\}$ (1)

Range $\{\text{all real } y\}$ (1)

Question 12.

(a) (i) $\sqrt{52} = 2\sqrt{13}$ (1) (either or)

(ii) $M_{AB} = \frac{3}{2}$ (1)

(iii) $y - y_1 = m(x - x_1)$
 $y - 4 = \frac{3}{2}(x - 6)$

$2y - 8 = 3(x - 6)$ (1)

$2y - 8 = 3x - 18$

$3x - 2y + 10 = 0$ (1)

(iv) $d_1 = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$

$d_1 = \frac{|(3 \times -1) + (-2 \times 3) + 10|}{\sqrt{3^2 + (-2)^2}}$ (1)

$d_1 = \frac{1}{\sqrt{13}}$ or 0.28 or $\frac{\sqrt{13}}{13}$ (1)

$(x - a)^2 + (y - b)^2 = r^2$

$(x - 0)^2 + (y - 2)^2 = 9$ (1)

$x^2 + (y - 2)^2 = 9$ (1)

$f(x) = \begin{cases} -2 & , x \leq 0 \\ x^3 & , x \geq 0 \end{cases}$

$f(-1) - f(2)$

$f(-1) = -2$

$f(2) = 2^3$ (1)

$= 8$

$f(-1) - f(2) = -2 - 8$

$= -10$ (1)

(d) (i) In $\triangle ACD$ and $\triangle ABE$,

$$\text{Let } AB = x$$

$$AC = 2x$$

$$AE = y$$

$$AD = 2y$$

$$\frac{AB}{AC} = \frac{x}{2x}$$

$$\frac{AB}{AC} = \frac{1}{2}$$

$$\frac{AE}{AD} = \frac{y}{2y}$$

$$\frac{AE}{AD} = \frac{1}{2}$$

$$\frac{AE}{AD} = \frac{1}{2}$$

$$\frac{AE}{AD} = \frac{1}{2}$$

① sides in proportion.

$\angle BAE = \angle CAD$ (common angle) ① angle.

$\therefore \triangle ACD \parallel \triangle ABE$

(2 sides in proportion and equal angle)

(ii) Since $\triangle ACD \parallel \triangle ABE$,
 $\angle AEB = \angle ADC$ (matching angles)

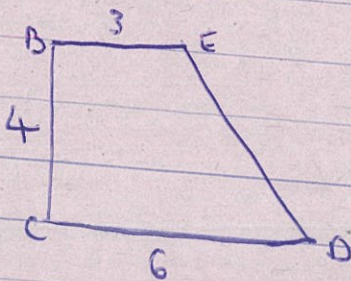
$\angle AEB$ and $\angle ADC$ are corresponding.

$\therefore BE \parallel CD$ (corresponding angles equal on parallel lines) ①

(iii) $A = \frac{h}{2} (a+b)$

$$A = \frac{4}{2} (3+6)$$

$$A = 18 \text{ cm}^2$$
 ①



Question 13

$$(a) M(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$M(2, 4) = \left(\frac{x+3}{2}, \frac{y+2}{2} \right)$$

$$\frac{x+3}{2} = 2$$

$$x+3 = 4$$

$$x = 1$$

$$\frac{y+2}{2} = 4$$

$$y+2 = 8$$

$$y = 6$$

$$\therefore C(1, 6)$$

$$(b)(i) f(x+h) = (x+h)^2 - 7(x+h) + 6$$

$$= x^2 + 2xh + h^2 - 7x - 7h + 6$$

either or. ①

$$(ii) f(x+h) - f(x) = x^2 + 2xh + h^2 - 7x - 7h + 6 - x^2 + 7x - 6$$

$$= 2xh + h^2 - 7h$$

①

$$(iii) \lim_{h \rightarrow 0} \frac{2xh + h^2 - 7h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 7)}{h}$$

$$\lim_{h \rightarrow 0} 2x + h - 7$$

$$\text{Sub } h = 0$$

$$= 2x - 7$$

①

$$\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5} =$$

$$= \lim_{x \rightarrow 5} \frac{(x-5)(x+5)}{x-5}$$

①

$$\lim_{x \rightarrow 5} x + 5$$

$$\text{Sub } x = 5$$

$$= 10$$

①

$$(d) (i) \frac{d}{dx} = 12x - 1$$

①

$$(ii) \frac{d}{dx} = \frac{3}{2} x^{-1/2}$$

$$\text{or } \frac{3}{2\sqrt{x}}$$

①

$$(iii) \frac{d}{dx} = 3(x^4 - 2x)^2 \times (4x^3 - 2)$$

$$= 3(4x^3 - 2)(x^4 - 2x)^2$$

$$(iv) y' = uv' + vu'$$

$$\text{let } u = x^2$$

$$u' = 2x$$

$$v = (x-4)^5$$

$$v' = 5(x-4)^4 \times 1$$

$$y' = 5x^2(x-4)^4 + 2x(x-4)^5$$

$$y' = x(x-4)^4(5x + 2(x-4))$$

$$y' = x(x-4)^4(5x + 2x - 8)$$

$$y' = x(x-4)^4(7x - 8)$$

① fully factorised.

$$(v) y' = \frac{vu' - uv'}{v^2}$$

$$\text{let } u = 2x$$

$$u' = 2$$

$$v = 7 - 3x$$

$$v' = -3$$

$$y' = \frac{2(7-3x) + 6x}{(7-3x)^2}$$

$$y' = \frac{14 - 6x + 6x}{(7-3x)^2}$$

$$y' = \frac{14}{(7-3x)^2}$$

① fully simplified.

Question 14.

(a) $y = 3x^2 - 2$

$$\frac{dy}{dx} = 6x$$

sub $x = 1$

$$\frac{dy}{dx} = 6$$

$$m_{\text{tangent}} = 6$$

$$m_{\text{normal}} = -\frac{1}{6} \quad (1)$$

If $x = 1$, $y = 3 \times (1)^2 - 2$
 $y = 1 \quad (1)$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = -\frac{1}{6}(x - 1)$$

$$6y - 6 = -x + 1$$

or

$$y = \frac{-x + 7}{6}$$

$$x + 6y - 7 = 0$$

(1)

b)(i) $\frac{\sin C}{c} = \frac{\sin B}{b}$

$$\frac{\sin C}{50} = \frac{\sin 122}{75}$$

(2) Finding 122° with reasoning.

$$\sin C = 50 \times \frac{\sin 122}{75} \quad (1)$$

(ii) $C = \sin^{-1}\left(\frac{50 \times \sin 122}{75}\right)$

$$C = 34^\circ$$

$$\angle CAB = 180 - 34 - 122 \quad (1) \text{ (angle sum triangle)}$$

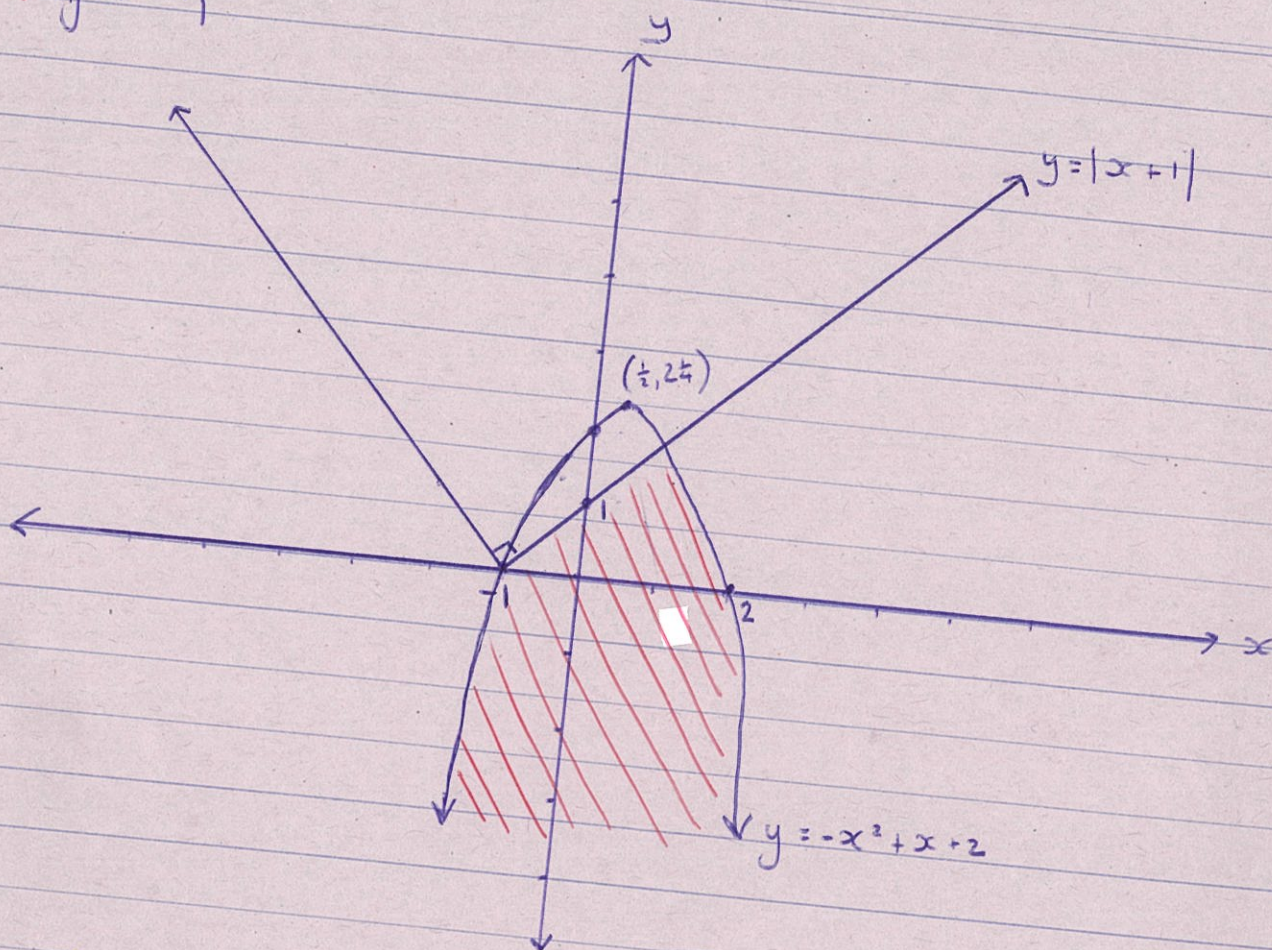
$$\angle CAB = 24^\circ$$

$$\therefore \text{Bearing} = 212 + 24$$

$$= 236^\circ$$

(1)

(c)(i) $y = |x+1|$



- ① x-y intercepts
- ② correct shape.

(ii) $y = -x^2 + x + 2$

x-intercept

$$0 = -x^2 + x + 2$$

$$x^2 - x - 2 = 0$$

$$(x+1)(x-2) = 0$$

$$x = -1, \quad x = 2.$$

$$(-1, 0) \quad (2, 0)$$

vertex

$$x = \frac{-b}{2a}$$

$$x = \frac{-1}{2 \times -1}$$

$$x = \frac{+1}{2}, \quad y = 2\frac{1}{4}$$

y-intercept.

$$y = 2.$$

$$(0, 2)$$

(iii) Test $(0, 0)$

$$0 < |0+1|$$

$$0 < 1$$

True

$$0 < 0+0+2$$

$$0 < 2$$

True